3.3.10 Aromatic chemistry (A-level only)

Aromatic chemistry takes benzene as an example of this type of molecule and looks at the structure of the benzene ring and its substitution reactions.

3.3.10.1 Bonding (A-level only)

Content	Opportunities for skills development
The nature of the bonding in a benzene ring, limited to planar structure and bond length intermediate between single and double. Delocalisation of p electrons makes benzene more stable than the theoretical molecule cyclohexa-1,3,5-triene. **Students should be able to:** • use thermochemical evidence from enthalpies of hydrogenation to account for this extra stability • explain why substitution reactions occur in preference to addition reactions.	

3.3.10.2 Electrophilic substitution (A-level only)

Content	Opportunities for skills development
Electrophilic attack on benzene rings results in substitution, limited to monosubstitutions. Nitration is an important step in synthesis, including the manufacture of explosives and formation of amines. Friedel–Crafts acylation reactions are also important steps in synthesis. **Students should be able to** outline the electrophilic substitution mechanisms of: • nitration, including the generation of the nitronium ion • acylation using $AlCl_3$ as a catalyst.	**AT b, d, g and h** **PS 2.1, 2.3 and 4.1** Students could carry out the preparation of methyl 3-nitrobenzoate by nitration of methyl benzoate, purification by recrystallisation and determination of melting point.